

ASSIGNMENT 1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

>> **Frontend** = what the user sees and interacts with (buttons, layout, colors, animations).

Ex: Making a "Buy Now" button look good and clickable on Amazon.

Backend = the logic and data handling that runs behind the scenes (server, database, business logic).

Ex: students' data is recorded in the college server.

Full-stack = it is the combination of frontend and backend both.

Ex: A dev who builds the whole Amazon checkout flow — UI + payment logic + database — solo.

2. Create a simple diagram showing how the client-server model works in web architecture.

>> **Client:** Sends a request (e.g. opens a website, clicks a button)

Server: Receives the request, processes it, sends back data

Client: Displays the response to the user.

3. Describe how a browser requests and displays a web page from a web server.

>> Type URL > DNS finds server IP > browser requests page > server responds with files > browser shows the page.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

>> **Code editor**- used to write code.

Web browser – to review our website.

Git/github - to make track of our project.

5. Explain what a web server is and give examples of commonly used servers.

>>A web server is software + hardware that stores website files and sends them to browsers when requested.

Examples: **Apache**

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

>>**Frontend developer**: Builds UI/UX .HTML, CSS, JS, what user sees and interacts with.

Backend developer: Builds server logic, APIs, handles business rules, connects to database.

Database administrator (DBA): Manages database design, security, backups, performance, data integrity.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

>>Download VS Code from web.

Open VS Code → go to Extensions

Install: Live Server, Prettier, HTML CSS Support, JavaScript (ES6)

Create a test .html file, right-click → "Open with Live Server" to check it works

8. Explain the difference between static and dynamic websites. Provide an example of each.

>>**Static**- websites which dont have any database but only frontend eg: simple portfolio website.

Dynamic- websites which have an database +backend eg: Amazon, Flipkart.

9. Research and list five web browsers. Explain how rendering engines differ between them.

>>Browser-Rendering, EngineChrome-Blink,Firefox-Gecko,Safari-WebKitEdge-Blink(Chromium-based),Opera-Blink

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

